BIOL3301 Marine Biology

Damian Lau

2025-2026 Semester 1

Introduction

This course begins with a discussion of marine biology, covering oceanography, shallow marine rocky subtidal systems, coral reef communities, polar communities, pelagic environments, intertidal communities, marine biodiversity, physiology of marine organisms and environmental changes, human impact on the sea and deep-sea communities. This course aims to introduce the whole marine system. Upon completion, students will be familiar with the mechanism and phenomenon in ocean.

Contents

1 Lecture 12 & 13 Thermal Marine Invertebrates & Fisheries Aquaculture

1. Metabolism
   - A process by which individual organisms acquire energy and materials from their environments
   - Use them for maintenance, differentiation, growth, reproduction
2. Metabolic Rate
   - $\uparrow$ Size, $\downarrow$ Ratio of Surface Area to Volume (Mass)
   - Metabolic rate is scale to mass$^{\frac{2}{3}}$
   - Metabolic Rate = Normalization Constant $\times$ Body Mass$^{\text{Scaling Coefficient}}$

$$\boxed{Q} = b_0 \boxed{M}^{b}$$

Figure 1: Metabolic Rate

3. Analysis Problem
   (a) Cannot use the same life stage
   $\Rightarrow$ E.g. Cannot use adults only
   $\because$ Adult have a large
   (b) Do not only looking at one species or one animal
   $\Rightarrow$ E.g. Do not use mammals only
   (c) Do not base on Basal or Routine or Standard metabolic rate (SMR)
   $\because$ Never appears in reality, even when the animal is resting $\because$ And, metabolism increase with activity in a lot of organisms
   $\Rightarrow$ E.g. Vertebrates
4. Metabolic Theory of Ecology = MTE
5. Oxygen and Capacity Limitation of Thermal Tolerance = OCLTT

6. Scaling Mechanisms : How a biological trait (E.g. Metabolic Rate) changes as the size (M = Body Mass) of an organism increases
   (a) West, Brown, Enquist (WBE) Model
      - Vascular Supply Network
        $\Rightarrow$ Measuring the reaching part of the body $=$ O_2 Rate
      - Delivery of non-storage nutrients : Oxygen
      - Resources Limitation
      - Enhance the metabolism
      - Used to measure the rate of O_2
        $\Rightarrow$ Not measuring the total surface area
      - Metabolic Rate : $M^{\frac{3}{4}}$
   (b) Dynamic Energy Budget (DEB) Model : Organisms maintain internal reserves (E.g. Proteins, Lipids, Carbohydrates)
      $\Rightarrow$ Used to fuel metabolism
      - Reverse Dynamics
      - Mobilization of storage nutrients : Proteins, Lipids, Carbohydrates
      - Basis for energy deliver in body
      - Energy Limitation
      - Seldom use this model
7. Metabolic Theory of Ecology : Metabolic Rate $=$ Normalization Constant $\times$ Body Mass$^{\text{Scaling Coefficient}}$ $\times\, e^{\frac{-E}{k\times \text{Temperature}}}$
   - Temperature $=$ Body Temperature
   - Need energy to regulate the body temperature of an organism
     $\Rightarrow$ Especially Endotherm
     $\Rightarrow$ A method to differentiate ectotherm and endotherm
   - Can check Metabolic Rate (Individuals), Development Rate (Individuals), Population Growth (Population), Mortality Rate (Population), Carbon Turnover (Community and Ecosystem), and Species Richness (Community and Ecosystem)

- Measuring the metabolic proxy, which is the oxygen rate
  $\Rightarrow$ Not really measuring the metabolic rate
  $\Rightarrow$ Related to body temperature of an organism
- Measuring an organism's performance and fitness
  $\Rightarrow$ E.g. How fast to do production when facing harmful condition

$$\boxed{Q} = b_0 \boxed{M}^b e^{-E/k\boxed{T}}$$

Figure 2: Metabolic Theory of Ecology

8. Thermal Biology and MTE
   - Metabolism : Understand relationship between organisms and environment
   - Proxy : Measuring Oxygen rate
   - Related to body temperature of an organism
   - related to a measure of performance or fitness
9. Body Temperature and Rates
   - Fitness = Metabolic Performance = Aerobic Scope
     $\Rightarrow$ Aerobic Scope = Ratio of Maximum Aerobic Metabolism : Standard Metabolic Rate (SMR)
   - Standard Metabolic Rate (SMR) : The maintenance cost of an organism
   - Maximum Aerobic Metabolic Rate (MMR) : The highest rate of oxygen consumption an animal can achieve during intense physical activity
   - Example of Fitness : Movement Speed, Reproductive Development, Feeding Rate, Digestion Rate
   - Optimal Body Temperature : While Reproduction

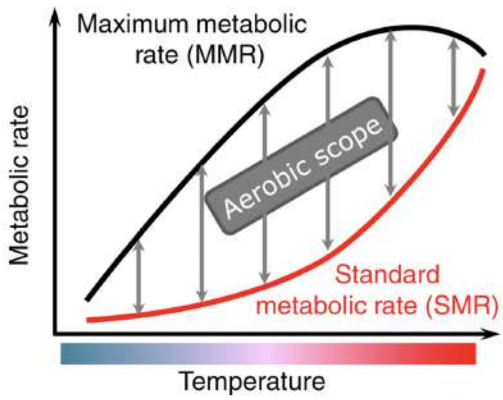


Figure 3: Aerobic Scope

- Optimal Temperature $\neq$ Peak of Maximum Metabolic Rate (MMR)
  $\because$ High temperature might cause protein damage (Denature)
- Lower Critical Limit Body Temperature (CT_{min}) $\neq$ die
  $\Rightarrow$ Only means the metabolic rate $\to 0 =$ Lowest normal limit of performance
  $\Rightarrow$ It can only maintain basic functions
- Tolerance Range : Temperature range between Lower Critical Limit Body Temperature (Minimum) and Upper Critical Limit Body Temperature (Maximum) (CT_{max})
  $\Rightarrow$ Animals can survive and function aerobically

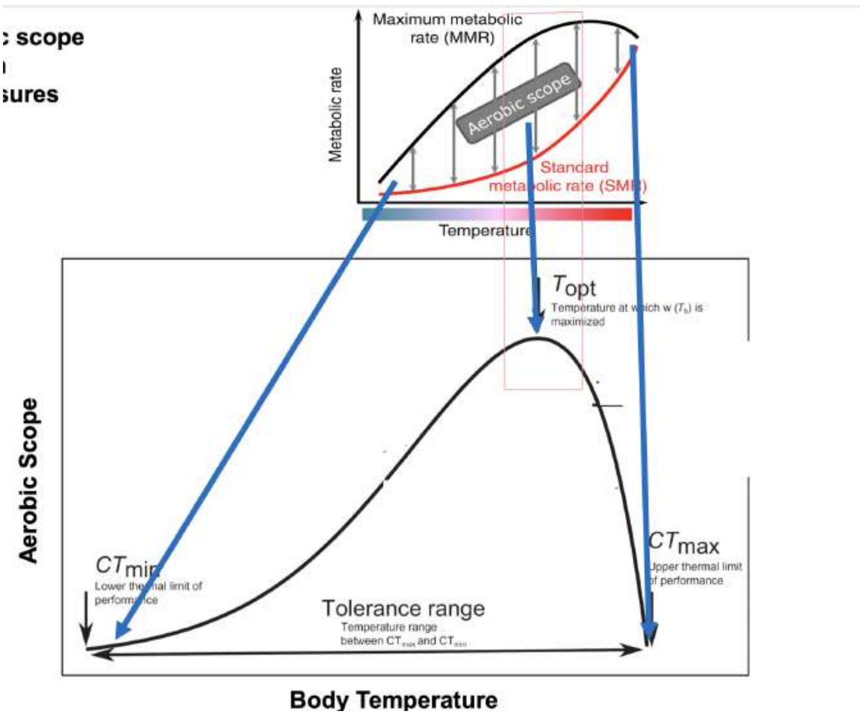


Figure 4: Metabolic Rate vs Temperature

- Climate change is changing environmental temperature

10. Physiology vs Climate Change
    - Performance = Potential Mortality + Species Range Shift
    - Oxygen and Capacity Limitation of Thermal Tolerance (OCLTT)
      - Can Predict Changes : Measure the changes in aerobic scope (Performance) under hotter temperature
      - Only based on Fast Moving Ectotherm
        $\Rightarrow$ E.g. Fish
    - OCLTT Mechanism
      - When temperature rises above optimal $\rightarrow$ The organism activates three types of acclimation responses to face it
      (a) Acclimation in functional capacity : Trying to adjust mitochondrial functions efficiency and energy consumption
      (b) Acclimation in protection : Use a Regulator Protein (HIF-1)

⇒ Turns on genes to enhance O_2 supply pathways
⇒ E.g. Making more red blood cells or Growing new blood vessels
⇒ Then, promotes metabolic depression to save oxygen
⇒ Result : ↑ Anaerobic Capacity ; ↑ O_2 Supply Pathways

(c) Acclimation in repair : Increasing Heat Shock Proteins (HSPs)
⇒ Refold damaged proteins and antioxidants to combat oxidative stress

– When temperature keep increase, the acclimation responses failed
  (a) The cardiorespiratory system (Heart & Gills / Lungs) can no longer meet the body's oxygen demand
  (b) Leads to Hypoxaemia (Low O_2 in blood) & CO_2 Accumulation
– Oxygen transportation failed → Tissues switch to emergency mode
  (a) Anaerobiosis : Anaerobic Respiration and produce lactate acid
  (b) Leads to Protein Denaturation and Hypoxia

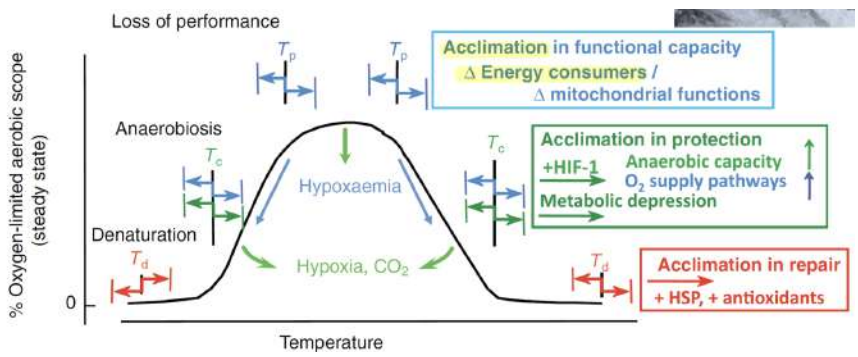


Figure 5: Loss of Performance

11. Thermal Biology of marine invertebrates (Move very slow)
    • Low metabolic rates

- No large changes in metabolism with activity
- Do not calculate the performance to measure the metabolic rate
- The highest temperature $\neq$ Optimal Temperature under aerobic scope
  $\Rightarrow$ Is the temperature of maximum metabolic rate
- Metabolism / Performance / Fitness : Metabolic Debt + Mortality (Long-term)

12. Metabolism and Performance are two different measures
    - Performance has decreased before metabolic rate reaches maximum
    - Differ measuring item between highly mobile vertebrates and invertebrates
13. Performance is different for different processes (Even in the same individual : Have different optimal temperature
    $\Rightarrow$ E.g. Movement, Digestion, Reproduction, Growth
    $\Rightarrow$ Impossible make everything to be optimal by keep adjusting our body temperature

Fisheries, Management and Aquaculture

1. 90% of world's fisheries are at or over maximum harvest capacity
2. Why Overfishing ?
   (a) Higher demand
   (b) Increase geographic and depth range of fisheries with better vessels and gears
   (c) Inaccurate estimation of stock
   $\Rightarrow$ Overestimation of sustainable fishing level
   (d) Roving Banditry
   (e) Illegal and destructive practices
3. Recruitment to biological population

$$N_{t+1} = N_t + B - M + I - E$$

   - N_{t+1} = Stock size at time $t+1$ year
   - N_t = Stock size at time t (Present Time)
   - B = Birth
   - M = Mortality : Natural, Fishery-induced
   - I = Immigration
   - E = Emigration
   - Stock Numbers : Proxy of stock size = Catch per unit effort (CPUE)
4. Potential Difficulties with stock estimation
   - Population is not homogeneous with respect to population parameter
   $\Rightarrow$ Adult vs Juveniles (Different)
   - Species are not equally distributed
   - Fishery often biased
   $\Rightarrow$ Only catch high valued fish
5. EI Niño and the Peruvian anchoveta

- Decrease in offshore trade winds
- Decrease in upwelling

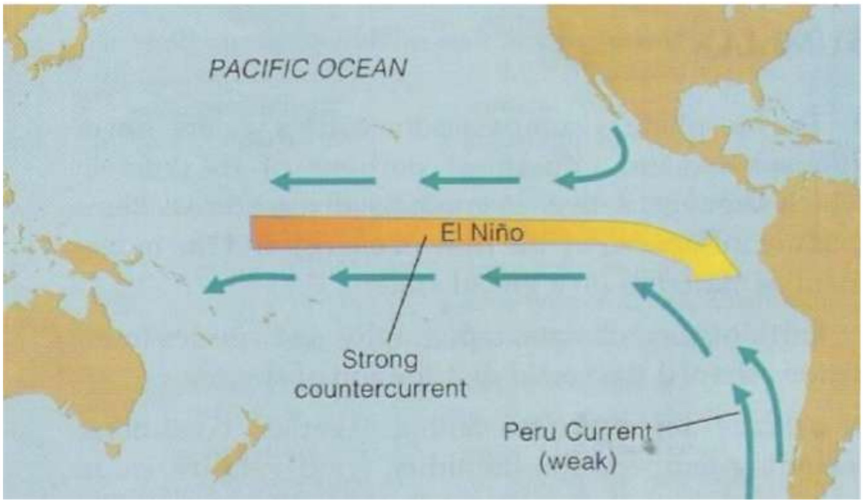


Figure 6: EI Niño

6. Habitat Damage Impact

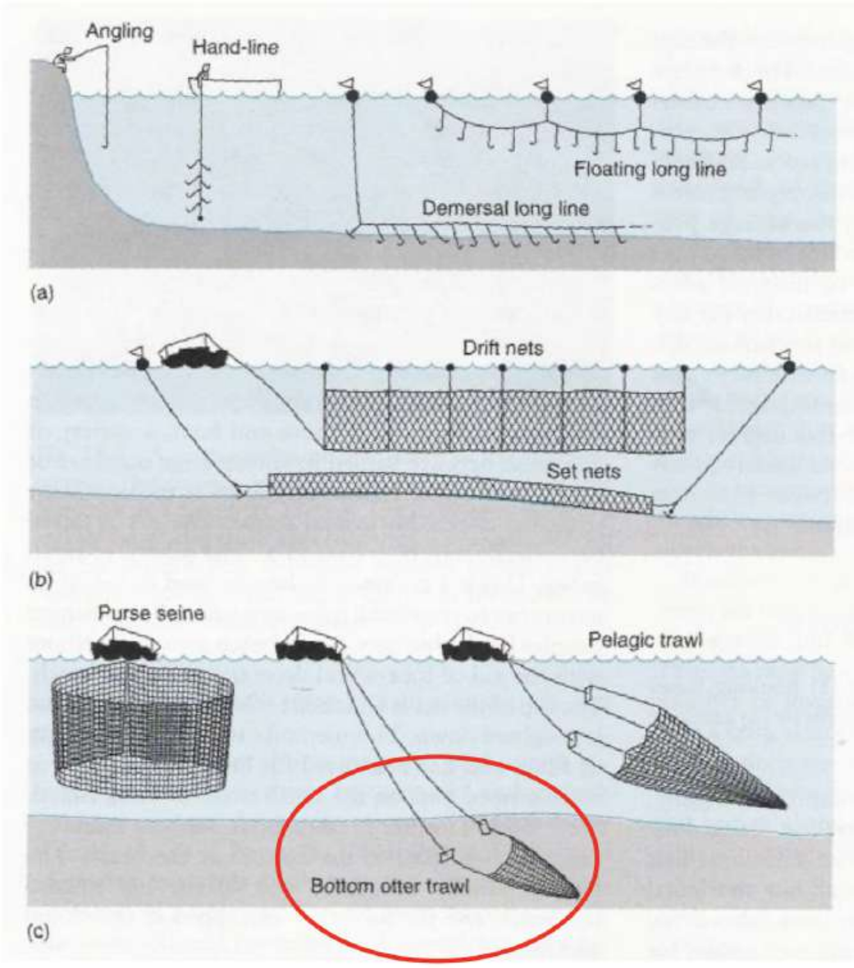


Figure 7: Fisheries Damage

7. Bycatch (Discards)
    - Low valued sea products
    - Mainly from prawn and finfish trawlers
    - Can also impact other fisheries
    - 40% of total catch
8. Types of Aquaculture
    (a) Extensive
        - Exclude Predators, Provide Space
        - E.g. Oyster, Mussels, Algae

(b) Semi-intensive
- Cage Aquaculture
- E.g. Tuna

(c) Intensive
- High Valued
- E.g. Salmon

9. Potential Advantages of Commercial Demand Aquaculture
   - Increase seafood production
   - Relax the pressure on wild fisheries
   - Economic benefits
     $\Rightarrow$ Community self-sufficiency and economic development
     $\Rightarrow$ Reduced cost of harvesting (cf. fishing)

10. Potential Disadvantages of Commercial Demand Aquaculture
    - Impacts near-shore environment
      $\Rightarrow$ E.g. Development, Removal of habitat, Pollution by effluent
    - May deplete wild stocks through
    - Capture of breeding individuals (Semi-intensive aquaculture)
    - Wild fishery species used for feed
      $\Rightarrow$ E.g. Anchovy
    - Risk of non-native species introductions (Invasive Species)
      $\Rightarrow$ E.g. Oyster Magallana Gigas

11. Sustainable Aquaculture
    - Eliminate the destruction of coastal habitats
      $\Rightarrow$ Create Aquaculture Ponds
    - Reduce reliance (Fish Powder) on fish meal as food for farmed species
      $\Rightarrow \uparrow$ Production of Herbivores $+ \downarrow$ Production of Carnivores (Food Source)
      $\because$ The fish powder are made from wild-caught fish

$\Rightarrow$ Make pressures to wild fish stocks, competing with marine predators and destabilizing marine food webs

- Introduce polyculture systems
  $\because$ Mimics natural ecosystem functions to increase efficiency
  + Reduce Waste + Enhance Resilience

12. Integrated Multi-Trophic Aquaculture (IMTA)
    - More Efficient, Little Waste, Greater Output
    - Have Complementary Biology and Roles
    - Have Commercial Value
    - Adaptable to Multiple Environments
    - Increased Productivity
      $\Rightarrow$ Kelp and Mussels can grow faster in salmon farms
    - Reduction of nutrient impacts on environment
      $\Rightarrow$ Nitrogen waste $\downarrow$ 20-70% by algae
      $\Rightarrow$ Mussels can filter particle waste and phytoplankton

13. Restorative Aquaculture : Partial ecosystem benefits of aquaculture
    $\Rightarrow$ Don't need any food, but can help to clean up the environment

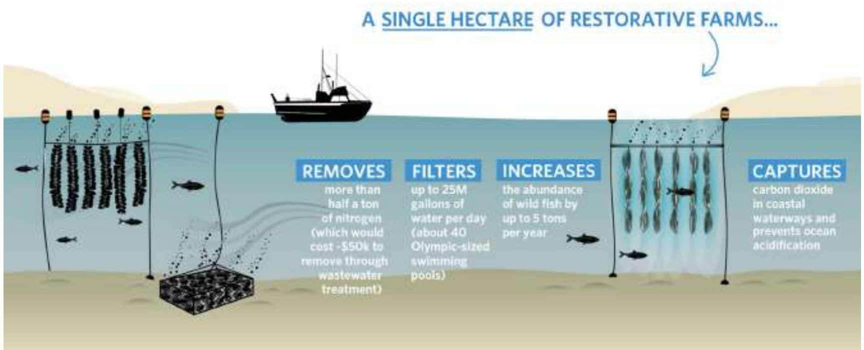


Figure 8: Restorative Aquaculture

2 Lecture 14 & 15 Marine Protected Area & Land Based Impact

1. Marine Protected Area (MPAs) : Designated areas in the ocean within which human activities and protected more stringently than other place

   - Why MPAs ?

     – Reason 1 : Fishing down the food chain : If catch all big fish, then people will start catch the small fish
     $\Rightarrow$ Birth rate of fish < Fishing rate

     (a) Firstly, large fish is caught by fishers
     $\Rightarrow \downarrow$ Number of Large Predators & $\uparrow$ Number of Smaller Fish

     (b) But, Smaller fish is then start to be caught by fisher since there is no more large fish
     $\Rightarrow \downarrow$ Number of Smaller Fish & $\uparrow$ Number of Large Fish
     $\because$ Fishers try to catch small fish, so large fish have a little bit increase

     (c) Eventually, both large and small fish decrease

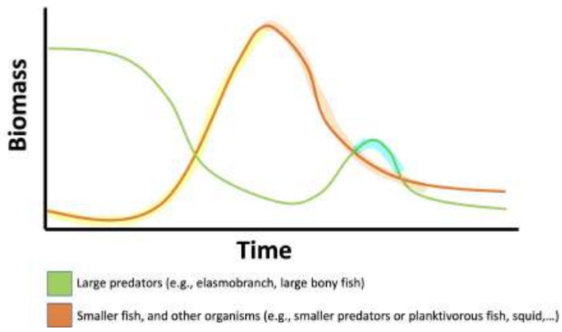


Figure 9: Large Fish vs Small Fish

     – Reason 2 : Endangered Animal Range is becoming larger and larger

     (a) Need to Protect the population of the large, overfished, vulnerable, high tropic-level species

- People usually underestimated fish age
  $\because$ Fishing deeper become easier
  $\Rightarrow$ We have better technology

2. Benefits of MPAs

- Increase in abundance and biomass of fishery targets
- Provide a baseline to assess population trends in non-protected areas
- (Indirect Effects) If we set MPAs, other Non MPAs might be more harmful
  $\because$ Number of fish catch remain unchanged and legal fishing place become smaller $\rightarrow$ Harmful to Non MPAs

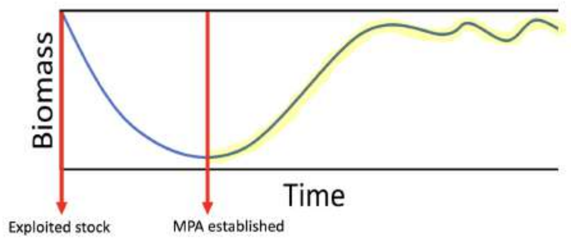


Figure 10: Biomass in MPAs

- $\uparrow$ Fish Density
  $\Rightarrow$ Larvae can grow because no predators (Fishing)
- $\uparrow$ Fish Size $\rightarrow$ $\uparrow$ Market Value
- $\uparrow$ Biomass
  $\Rightarrow$ Better breeding populations in MPAs

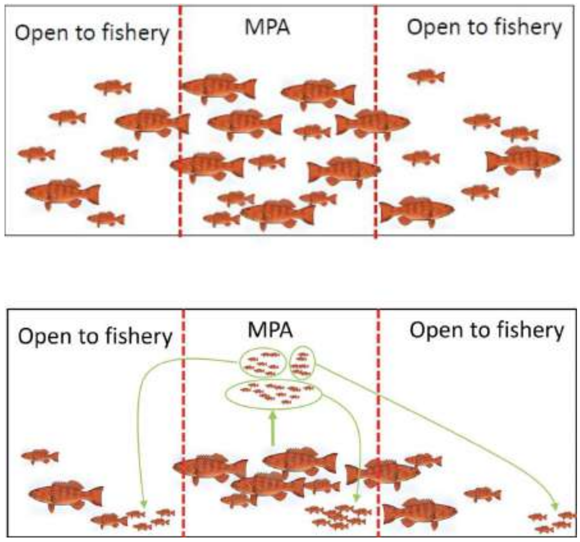


Figure 11: Indirect Effects on MPAs

3. Example of MPAs : Seasonal Harvest
4. Problems around MPAs
   - Lack of systematic planning and standardized framework
     ⇒ Cannot make sure there are no any illegal fishing in the MPAs
   - No adequate funding
   - No appropriate zoning
   - No long-term monitoring
   - Inadequate and absent enforcement
   - Solution 1 : Set MPA → Enforcement → Monitoring
   - Solution 2 : Has a hierarchy of different allowed or banned uses
     ⇒ From restricting only some activities, to complete exclusion
5. Only full and high levels of protection result in migration or adaptation make benefits
   ⇒ Lower levels of protection has no benefits fo climate mitigation

6. Climate Pathways are positively correlated with MPA age

Land Based Impact

1. Source of inputs
   - Non-point = We don't know the source
     - Agricultural Runoff
     - Urban Seapage
     - Land Activities that generate contamination
       ⇒ Logging, Construction and Development
   - May be more intermittent in outflow
   - Come from large areas of land
   - Difficult to measure and regulate
2. Sources of Chemical Inputs
   - Point = We know the source
     - Wastewater Effluent
     - Runoff from waste, construction sites
     - Runoff from feedlots
     - Runoff from mines, industrial sites
     - Storm water outfalls
   - Often continuous or predictable outflow
   - Simple to measure and regualte
3. A large number of "small points" sources make up a non-point source

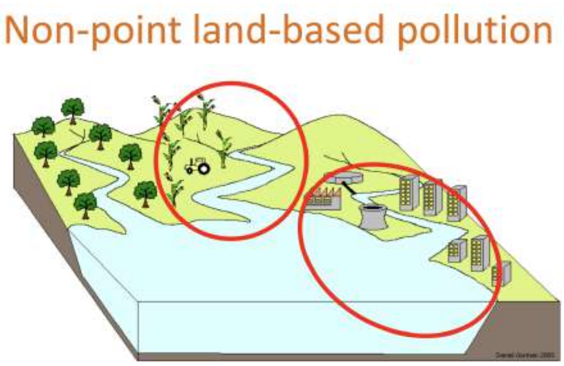


Figure 12: Non-point Land-based Pollution

4. Sedimentation
    - Increase erosion on land $\rightarrow$ Enhancing sedimentation on coasts
    - Construction in marine water
      $\Rightarrow$ E.g. Reclamation for land
    - Used to be a natural process
    - Sediments impact many communities through burial
    - Sediment smothers organisms
      $\Rightarrow$ Stops coral feeding
      $\Rightarrow$ Leads to death
    - Can turn 'hard-bottom' communities into 'soft-bottom' communities (Benthic Environment)
      $\Rightarrow$ E.g. Coral Reefs and Kelp Forests
    - A thin layer of sediments can already block recovery
    - Increase turbidity + Reduce light intensity
      $\Rightarrow$ Reduce Photosynthesis
    - Movement of sediments through wave action is able to hurt the sensitive living tissue
      $\Rightarrow$ E.g. The feeding tentacles of coral polyps
5. Eutrophication
    - Eutrophication : Excessive Nutrient Loading
    - Nitrogen : Limiting Nutrient in ocean
      $\Rightarrow$ Fixed by fertilizer for Agriculture, Animal Waste, Industrial Waste
    - But it will transported to the coast in river
      $\Rightarrow$ Most of them is not absorbed
    - microbial decomposition of nutrients will cause hypoxic / anoxic zones
      $\Rightarrow$ Kill most benthic life
    - Solution : Stop the eutrophication
6. Eutrophication Causes

- Cause community shift
- Coral Reef : High nutrition → Algae Developed → Block lights → Symbiotic bacteria cannot do photosynthesis → Coral Reef smothered and killed by dense algal mats
- Kelp Forest : Replaced by fast growing algal turf
- Increase frequency and duration of phytoplankton and microbial blooms
  ⇒ Blooms are toxic → Kill fish & shellfish

7. Eutrophication Consequence
   - Increase incidence of fish kills
   - Reductions in harvestable fish and shellfish
   - May lead to decreased ecosystem resilience and recovery capacity
   - Hypoxia and dead zone by algae
   - Potential human health impacts
8. The three R's shift : Resistance, resilience, Regime Shifts
   (a) Resistance
       - Ability of system to absorb disturbance
       - Maintain functions and structure
   (b) Resilience
       - Ability of system to bounce back to its original structure and function by following the disturbance
   (c) Multiple stressors can reduce both Resistance and Resilience
       → Regime Shifts
       ⇒ E.g. Eutrophication, Harvest, Climate Change
       ⇒ Can happen at the same time and in the same locations
       ⇒ They can interact
9. Catastrophic System Shift
   - Ball and Cup Model : Resilience and Resistance of ecosystems
     ⇒ Shift to the new, degraded state

- Impacted State : Heavy Eutrophication = The ball easily shift right = Shift to an unhealthy state
- Natural State : Little Eutrophication = The ball stay at left in the healthy zone
- **Shift to right : Very Easy ; Shift to left : Very Hard = Hysteresis**

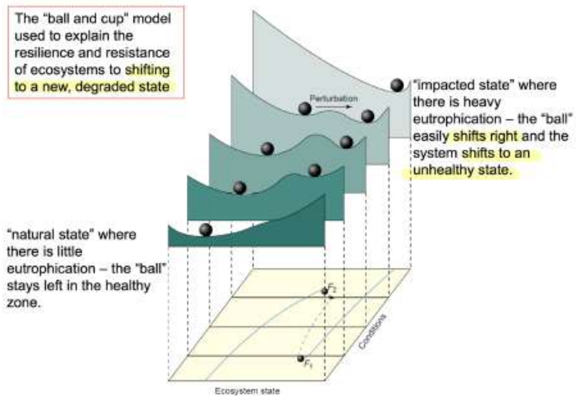


Figure 13: Catastrophic System Shift

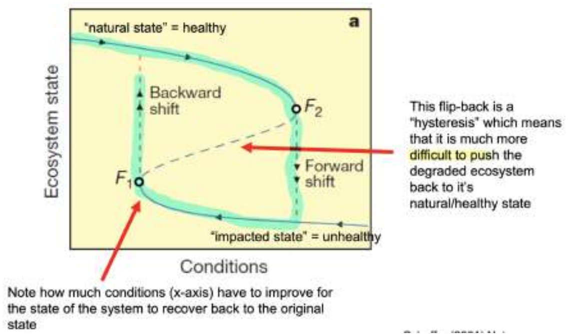


Figure 14: Shifting Curve

10. Major Seagrass Species : Posidonia

- Exceed nutrients (Slow Release Fertilizer)
  ⇒ Simulate eutrophication to patches of seagrass
- Overgrowth by fast growing algae when fertilizer is added
- Seagrass are not photosynthesis well and died

11. Great Pacific Garbage Patch : Plastic
    $\Rightarrow$ The most harm plastic : Synthetic Fibre
    $\Rightarrow$ Microplastics might enter the entire microplastic

3 Lecture 16 & 17 Ocean Heating & Ocean Acidification

1. Biological Effects of Ocean Warming
   - Upper Temperature Limits : Species tends to have Large Populations = Most Productive, Grow Best, Reproduce Fastest
   - More easy to leads to physiological stress and mortality
   - Tropical : Very stable temperature habitat
   - Polar : Narrow, Need to move somewhere else (Too hot)
2. Two Major Effects of Ocean Warming
   (a) Increase Stress to individual : Decrease productivity and leads to death
      - Coral Bleaching : A sign of extreme thermal stress in coral
        $\Rightarrow$ Energetic costs become too high $\rightarrow$ Expelling the symbiotic bacteria ***Symbiodinium*** $\rightarrow$ Become white
      - If the temperature doesn't reduce, corals cannot reinoculated with the symbiotic bacteria within a short period
        $\Rightarrow$ Leads to mortality
        $\because$ They have no energetic intake since energy is gained from the photosynthesis of the bacteria
      - Degree Heating Days (DHD) : Used to predict coral stress and threshold for bleaching
        $\Rightarrow$ **Predict Marine Heat Wave**

   (b) Species and Population Range Shifts
   $\Rightarrow$ Contractions / Shifts and Expansions
      - Black Spined Sea Urchin creates barrens
        $\because$ They eat all algae from the rocks
        $\Rightarrow$ Removes habitat from species $\rightarrow$ Reduce the productivity of the entire system
      - Ecological Devastation : Reduce habitat structure
3. Some species can adapt the warming : **Diatoms**
   $\Rightarrow$ Significant in different extremum

4. Marine Heatwaves : 90% of general SST variation
5. Heat Spike : $<$ 5 consecutive days

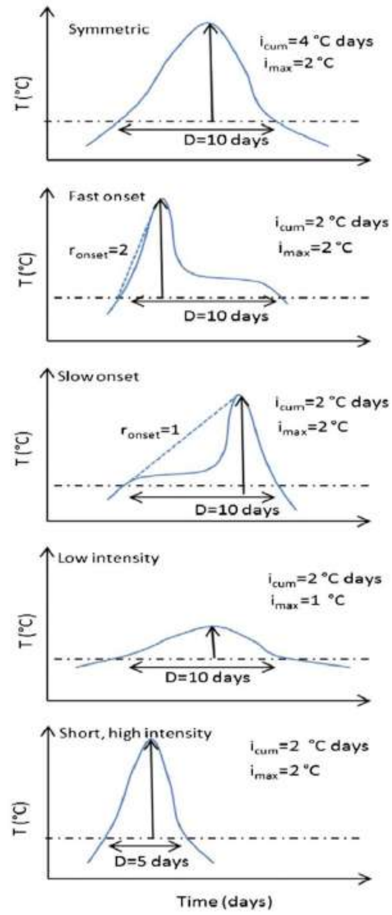


Figure 15: Different kind of Heat Waves

6. Species cannot responses the rapid global change quickly
7. Population Size in Marine Heatwaves
   ⇒ Species that cannot against heatwave will die
   ⇒ In long term, that gene will disappear
   ⇒ Reduce Genetic Diversity

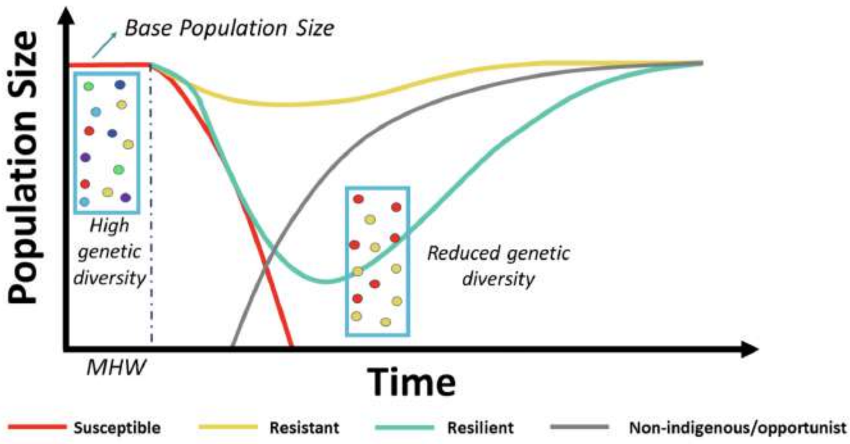


Figure 16: Population under marine heatwave

8. **Direct Effects** of Marine Heatwave on Seaweed
   - Local extinction
   - Range shift
   - Declined abundance
   - Tissue bleaching
   - Shifted competitiveness due to increased abundance
9. **Indirect effects** of Marine Heatwave on Seaweed
   - Increased grazing pressure
   - Increased abundance of invasive species
10. Effects between kelp forest and turf algae under marine heatwaves
    - Change from temperature kelp forests to turf dominated
      $\Rightarrow$ Not reversible
    - Even the water temperature has returned to normal, the tropical fish currently inhabiting the reestablishment of kelp forest
11. **The more and longer fossil fuels are burned, The more we force the climate system away from natural cycles**
    $\Rightarrow$ The higher the CO_2 concentration, The larger the negative effects of climate change will be

12. Change earlier $\rightarrow$ Environment can exponentially better

Ocean Acidification

1. Ocean Absorb : 30-40% CO_2 in atmosphere $\rightarrow$ Form Carbonic Acid (Ch.1)

2. Organism with calcium carbonate shells (E.g. Molluscs and Corals) are more difficult and energetically expensive to form shells
   $\because$ The saturation sate of carbonate is reduced
   $\Rightarrow$ Difficult to get from the water

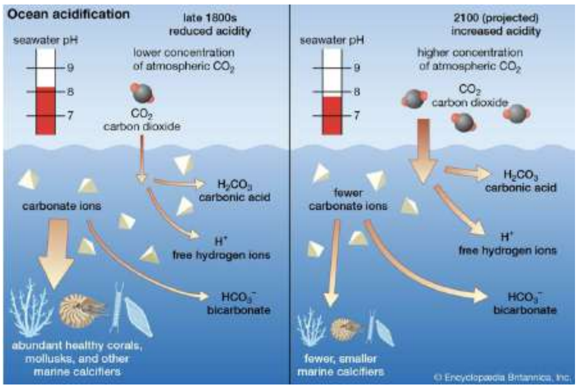


Figure 17: Ocean CO_2 Absorption

3. $\uparrow$ Atmospheric CO_2 Concentration $\rightarrow \downarrow$ Ocean pH (Acidic) $+ \downarrow$ Carbonate Saturation (Carbonate ion Concentration)
   $\Rightarrow$ This is followed by the formula

$$CO_2 + H_2O \longrightarrow H^+ + HCO_3^-$$

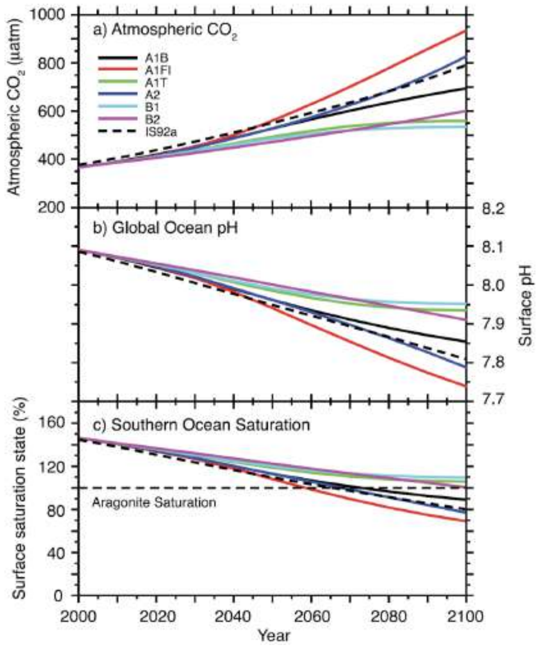


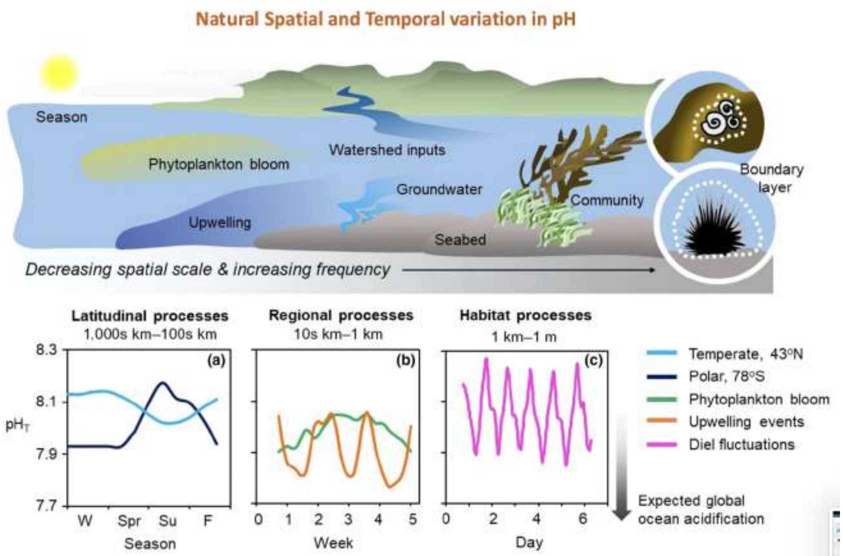
Figure 18: Atmosphere CO_2 vs pH vs Carbonate Saturation

4. Natural Spatial & Temporal Variation in pH
   (a) Latitudinal Processes
      - Spatial Scale : Ocean basin or regional
      - Time Scale : Seasonal
      - Key Drivers : Seasonal Temperature
        $\Rightarrow$ pH is temperature-dependent
        $\Rightarrow$ Colder water holds more $CO_2 \rightarrow \downarrow$ pH
   (b) Regional Processes
      - Spatial Scale : Coastal Zone, Bay, Estuary
      - Time Scale : Weekly
      - key Driver : Day Night Cycle : Photosynthesis during the day removes CO_2 (pH Rises)
        $\Rightarrow$ Respiration at night releases CO_2 (pH Falls)
      - Phytoplankton Blooms in Spring and Summer
        $\Rightarrow$ When phytoplankton bloom, they photosynthesize $\rightarrow$ consuming CO_2 and $\uparrow$ pH

- When the bloom dies or decays $\rightarrow$ respiration releases $CO_2 \rightarrow \downarrow$ pH
- Upwelling Effects : Bring Deep, Cold, Rich CO_2 (Low pH) water to the surface
  $\Rightarrow$ Sudden drops in coastal pH

(c) Habitat Processes

- Spatial Scale : Microhabitats
  $\Rightarrow$ E.g. Coral Reef, Seagrass Patch
- Time Scale : Daily
- Key Driver : Community Metabolism (Diet Fluctuation)



5. **Acidification is not homogeneously spatially or temporally**
6. Ocean Acidification Impacts

- Surface pH decrease 0.1 units over past century
- Coral Cores
  - Steady drop
  - pH $= -\log(H^+)$, where H^+ is positive charged $=$ Acidic
  - Small changes in pH have great impact on the carbonate system
- Lower pH

  - Carbonates undersaturated in solution
  - Calcium carbonate ($CaCO_3$) skeleton may dissolve and hard to form new skelecton
- Fluctuating pH can be more energetically and physiologically costly
  ⇒ Including growth / Reproductive output

7. Effects on marine fish
   - Acidification cause negative effects on several physiological aspects
     ⇒ Especially for the larvae fish in growth and survival
   - If parents have been adapt in high pCO_2 condition, offspring body size can adapt this extreme condition
   - Can affect the fish behaviour : Chemical Alarm Cues (CAC)
     ⇒ A chemical Receptor
   - Increased CO_2 concentrations can impair (Damage) behavior

8. Environmental and Evolutionary history drive response of Copepods
   - Organism from high CO_2 area : Less respond to pH change
   - Copepods are mainly affected by very low pH (Close to 7 or below)
   - Closely related species can respond similarly
     ⇒ But it can also respond very different

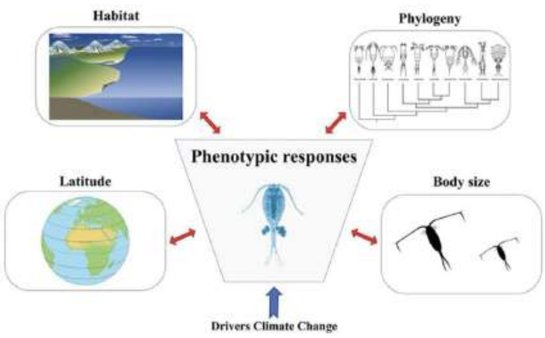
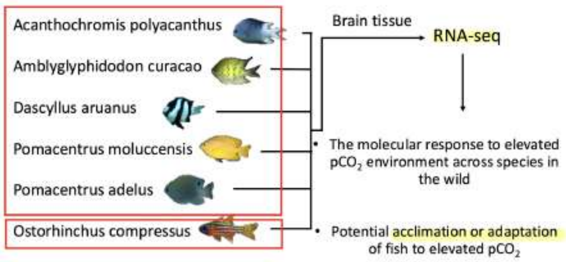


Figure 19: phenotype responses

9. CO_2 Seeps Example
   (a) White Island in New Zealand
   (b) Papua New Guinea
   (c) Italy

Figure 20: CO_2 Seeps

10. Density Compensation : Achieved by population bloom of small acidification-tolerant animal groups
11. The trophic structure of invertebrate communities shifted towards fewer trophic groups
    ⇒ Generalist (Omnivorous) species became dominant
    ⇒ Ocean acidification may lead to food web simplification
12. Natural CO_2 Seep in Papua New Guinea



13. CO_2 Seep : Variability & Adaptive Potential
    - Different gene expression patterns exist in the brains of fish living in areas with high CO_2 leakage
    - Have Adaptive Defense Mechanisms
    - Balancing selection : Provide species with the potential to adapt

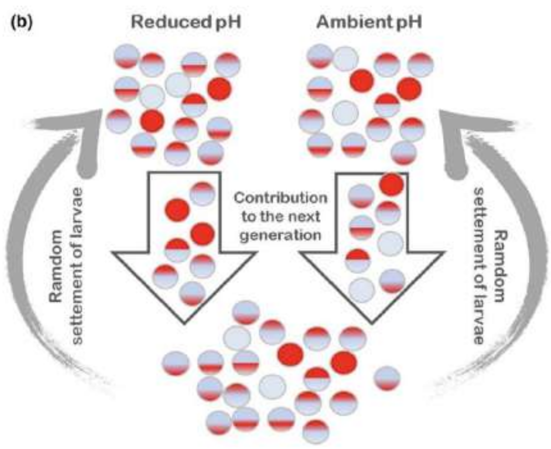


Figure 21: Variation & Adaptive Potential

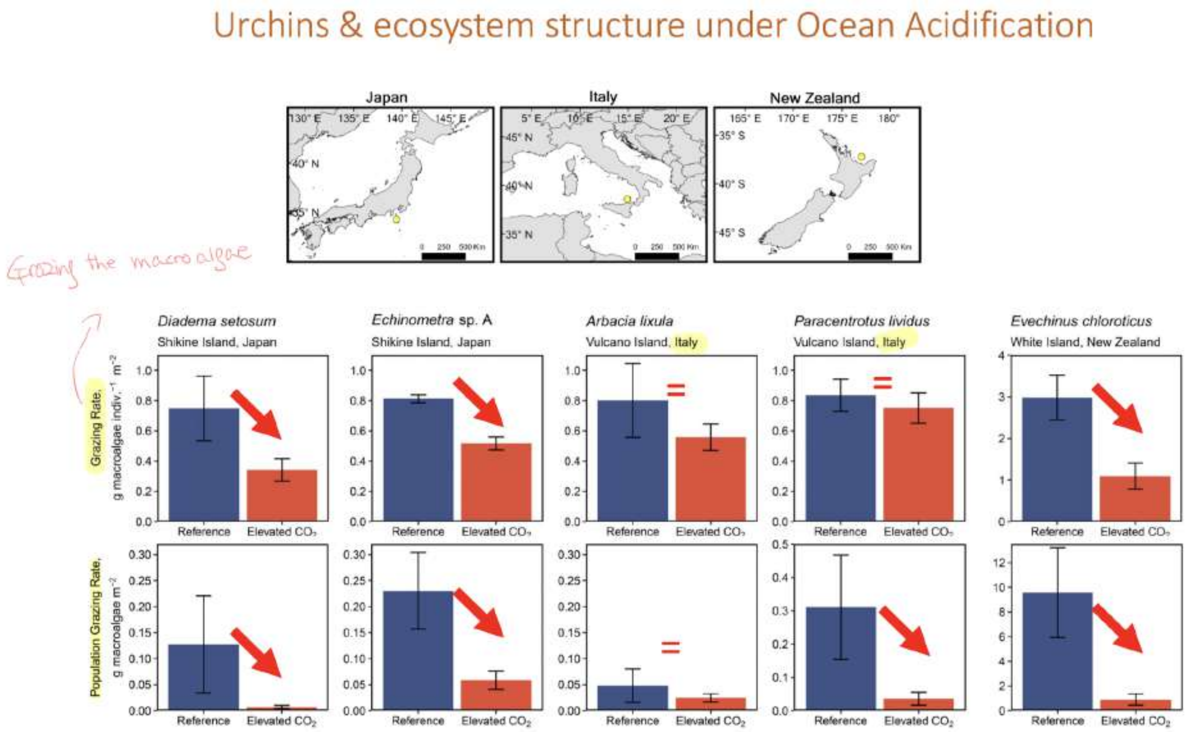


Figure 22: Urchin & Ecosystem under Ocean Acidification

4 Lecture 18 Deep Sea Environment

1. Aphotic Zone
   - Photic Zone : Light is present
     ⇒ Euphotic Zone : 0-200m
     ⇒ Disphotic Zone : 200-1000m
   - Aphotic Zone : Light is absent
     ⇒ Bathyal and Bathypelagic Zone : 1000-4000m
     ⇒ Abyssal and Abyssal-pelagic Zone : 4000-6000m
     ⇒ Hadal and Hadalpelagic Zone : 6000-10000m
   - Temperature in Aphotic zone $<$ Surface Temperature $= 0°C - 10°C$
   - Salinity in Aphotic Zone $>$ Surface Area
   - Abyssal plains cover 60% of earth surface
     ⇒ Abyssal plains = Flat + Deep Ocean Floor Regions

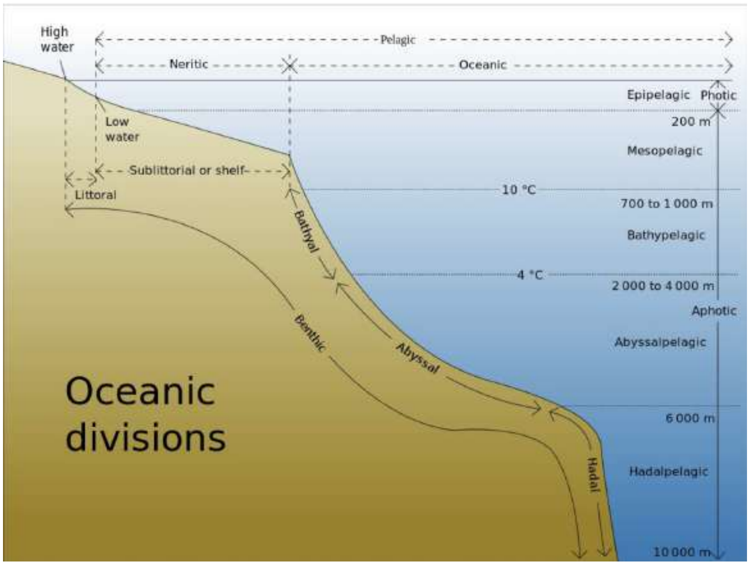


Figure 23: Ocean Division

2. General Characteristics
   - Environmental condition : Homogeneous

- No photosynthetic Productivity
  $\Rightarrow$ Food supply depends on productivity from chemical synthesis and above communities
- Organic matter is consumed in water column
  $\Rightarrow$ Hard to reach to deep sea
  $\Rightarrow$ Remaining material is tougher + No nutritional value
- Productivity, light, temperature, salinity : Highly variable in surface and coastal area (Base on seasonality and input form land)
  $\Rightarrow$ Stable at depth
- Some deep sea area have boost of organic matter
  $\because$ There are booms of productivity at the surface
  $\Rightarrow$ Deep sea productivity is depending on the shallow sea food productivity

3. Biodiversity with depth

- Decrease with depth
- Density of animal is very low in general
- Environmental conditions : Homogeneous
- Contain the highest biodiversity at intermediate depth
- Some part have relatively high density
  $\Rightarrow$ Hydrothermal vents, deep sea corals
  $\Rightarrow$ E.g. Invertebrates (Deep sea corals)
- Limit Human Impact
- Most Pristine Ecosystem

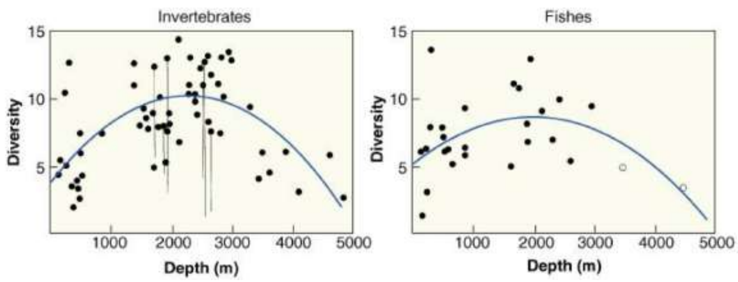


Figure 24: Biodiversity in Depth

4. Shallow Deep Comparison

   (a) Productivity : Deep Sea = Low ; Shallow Water = High
   ⇒ Deep Sea is more sensitive

   (b) Biomass : Deep Sea = Low ; Shallow Water = High
   ⇒ Deep Sea is more sensitive

   (c) Physical Energy : Deep Sea = Low ; Shallow Water = High
   ⇒ Deep Sea is more sensitive

   (d) Species Diversity : Deep Sea = High ; Shallow Water = Moderate and Variable
   ⇒ Deep Sea is more sensitive

   (e) Size of habitat : Deep Sea = Large and Contiguous ; Shallow Water = Moderate and Non Contiguous
   ⇒ Deep Sea is more robust

   (f) Species Distribution : Deep Sea = No Evidence ; Shallow Water = Narrow and Variable
   ⇒ No Conclusion

5. Adaptation to life at depth

   - Piezolytes : Change in fluidity of membrane
   ⇒ Stabilize proteins and enzymes against the immense pressures of the deep ocean
   - Absence of air filled sacs
   ⇒ Need to deal with the high pressure
   - Stabilization of compound (Proteins) and enzyme
   - Organisms move very slow
   - Slow Growth + Slow Metabolic Rates
   - Less muscle
   - Most species are blind / can only distinguish light from darkness
   - Red colour
   ⇒ Red light wavelength is not visible at depth
   - Some large organisms consume food sporadically (Weeks)

- Adaptation of consuming food
  - Larger jaws to eat the largest preys
  - Large stomachs to take as much food as possible
  - Very long intestine to abosorb as much nutrients as possible
- Staying close to mates
  ⇒ E.g. Parasitic males
- Bioluminescence

6. Bioluminescence in aphotic zone
   - Strategy to find mates, attract prey and deter predators
   - Produce by photophores
   - Can also release when threatened
   - Hatchetfish can also use bioluminescence for camouflage
     ⇒ Can only seen form the bottom
     ⇒ Convergent Evolution : Evolved independently multiple times in unrelated species
   - Mechanism of Bioluminescence
     (a) Luciferin : Emit light when it is oxidized
         ⇒ Catalyzed by the enzyme Luciferase
     (b) Luciferin have many different structure, depending on the organism
         ⇒ E.g. Present in fungi and dinoflagellates
     (c) Different luciferin can give different colours
     (d) Can produce this compound by themselves or by symbiotic bacteria

7. Microbial Activity
   - Very low POC
   - Deep sediments are almost no organic matter
   - Biomass of bacteria is low at depth
   - Compare to continental shelf, microbial activity is very low

- Uptake oxygen : Very slow rate than bacteria at the surface
- Slow Growth (Very long generation time)
- The crust can have seeps of hydrogen sulfide, methane or hydrogen
  $\Rightarrow$ Have high concentration of microbes
- Do not depend on the decomposition of organic matter generated by photosynthesis
- Chemo-autotrophs : Primary producers that can make energy breaking chemical bonds
- Can form very visible mats and aggregations

8. Deep Sea Coral Mounds
   - Can be found in deep areas (100-1000m)
   - Provide a complex 3D structure for many species
   - Heterotrophs and no photosynthetic symbionts
     $\Rightarrow$ Feed on suspended particles
   - Grow slowly and very long lived
   - Can trap organic matter
   - Problem : Human catch it by trawling for jewelry
9. Deep Sea Corals : Aragonite Saturation
   - Secrete calcium carbonate in the form of aragonite
   - Aragonite Saturation Horizon (ASH) : Maximum depth of aragonite can be synthesized
   - Calcite is not affected
   - Secretion : Driven by pressure, water temperature, CO_2 buildup from biological activity and absence of gas exchange with atmosphere
   - Usually in Atlantic 2000m depth and Pacific 50-600m depth
   - As acidification occurs, deep-sea limestone layers (ASH) will become shallower
     $\Rightarrow$ Changing deep-sea biological communities

10. Seamounts
    - Volcanic origin
    - Most of them are endemism (Only occurs in unique area)
    - Have hard substrates
      ⇒ Current directly impact the summit impeding accumulation of sediments
    - Local Upwellings : Deflect (Change) currents stirring (direction) and deflecting water
      ⇒ Accumulate nutrients and POC (Organic Carbon)
      ⇒ Attracts organisms
11. Whales and large fish falls
    - Occurs when whales and large fishes die
    - Carcasses can fall to the bottom
      ⇒ Bringing large amount of food for short periods of time (10 Decades)
    - Four stages of decomposition
      (a) Scavenger Stage : Large scavengers eat meat part
          ⇒ E.g. Fish, Sharks, Crabs
      (b) Enriched Sediment Stage : Invertebrates living on left-over flesh, ligaments, tissues + The enriched sediments from decomposing organic material
      (c) Sulfide stage : Organisms that rely on hydrogen sulfide
          ⇒ From anaerobic bacteria living in anoxic sediment
      (d) Reef stage : The remaining bones give support for benthic organisms
12. Hydrothermal Vents
    - Found in oceanic ridges (Lots of volcanic activity = Can heat water)
    - Hot water can dissolve metals and sulfide emanate from the crust

- Water does not boil
  ∵ Deep sea have intensive pressure
- Metals and chemicals go back to the ocean
- Surrounding water is cold
  ∵ High Heat Capacity → Strong thermal gradient

  (a) White Smokers : Farther away from the heat source, emanate barium , calcium or silicon

  (b) Black Smokers : Emit sulfide, iron

  (c) Black Smokers : Vents are formed by metal precipitation and accumulation
  ⇒ Also contain higher temperature

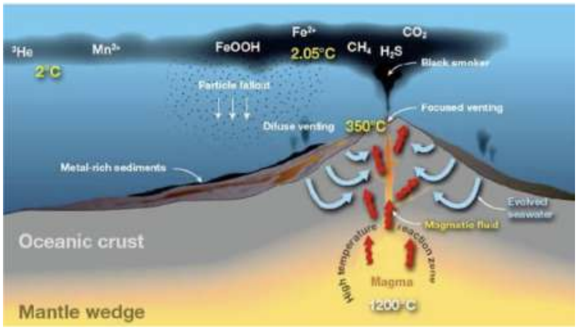


Figure 25: Hydrothermal Vents

13. Biodiversity of Hydrothermal Vents

- Primary Producer are Chemotrophs : Derive energy using a final electron acceptor in respiratory chain
  ⇒ Not the oxygen → ATP one
- Archea and bacteria can use iron, sulfur and carbon as final electron acceptor
- Produce hydrogen sulfide and methane as biproduct
- Can move freely forming large mats on rocky surface or form symbiotic relationships with other organisms
- Green Sulfur Bacteria : Have Photosynthetic Pigments
  ⇒ Generate energy form the geothermal radiation (Not Light)
- Directly grazed by crabs, shrimps, limpets and bivalves

14. Vestimentifera Worms
    - Foundation species
    - Facilitates the colonization of other groups
    - Lack a gut and digestive system
      ⇒ Depend on symbiotic bacteria
    - Trophosome : Organ with dense concentration chambers
      ⇒ Contain sulfide-oxidizing bacteria
    - Specialize hemoglobin : Binds to O_2 and Sulfide (To feed the bacteria)
    - Diagram : Collect H_2S through exposed plume + Bring the nutrients to the trophosome
    - Selective symbionts are not transferred from parents directly
15. Yeti Crabs
    - Have the capacity to eat mollusks and other crabs
    - Have structures similar to hairs
      ⇒ Accumulate symbiotic bacteria
    - Directly feed on the bacteria that grows on their body
16. Hydrothermal Vents : Ephemeral (Temporary)
    - Hot water source are temporary
    - Gastropods dominate in early colonization
    - Vestimentiferans dominate while vents are active
    - Cool Down : Bivalves become common (Dominated)
    - They disperse by larvae goes all the way up to the surface
    - Vents can quick replenishment from other sources after a volcanic eruption
17. Cold Seeps
    - Fissures in the crust
    - Sulfide, Salts, Methane can emanate

- Released water temperature is similar to the surrounding

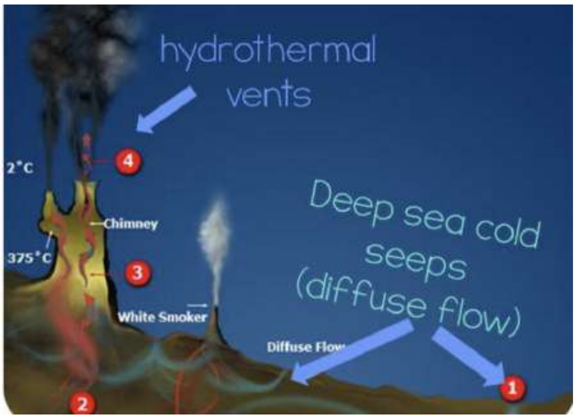


Figure 26: Cold Seeps

- Food Web : Based on chemo-autotrophs and free living and symbiotic
- Some areas will become brine pools (Salty Pool)
  $\Rightarrow$ Accumulate high concentration of salts, higher density than other areas
- Only some bacteria can live in

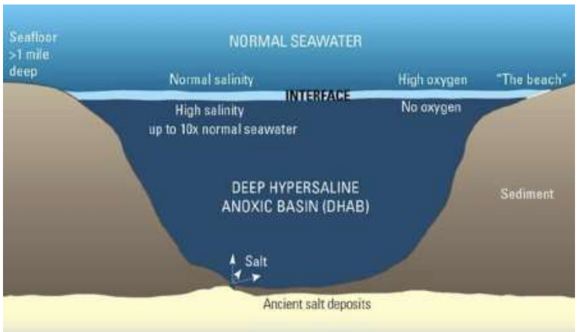


Figure 27: Brine Pool

18. Threats to the deep marine environment

- Oil and Gas Industry
- Fishing
- Trawling

- Deep Sea Mining
  - Formation of hydrothermal vents leads to the accumulation of large metal deposits
    $\Rightarrow$ E.g. Gold, Silver, Copper, Zinc
  - Include necessary electronic components
    $\Rightarrow$ E.g. Cobalt for cellphone
  - Do not know the capacity of recovering if we mining
    $\because$ The ephemeral nature of hydrothermal vents
  - Vent depend on other species
    $\therefore$ Destroy vents $\rightarrow$ Multiple taxa disappeared

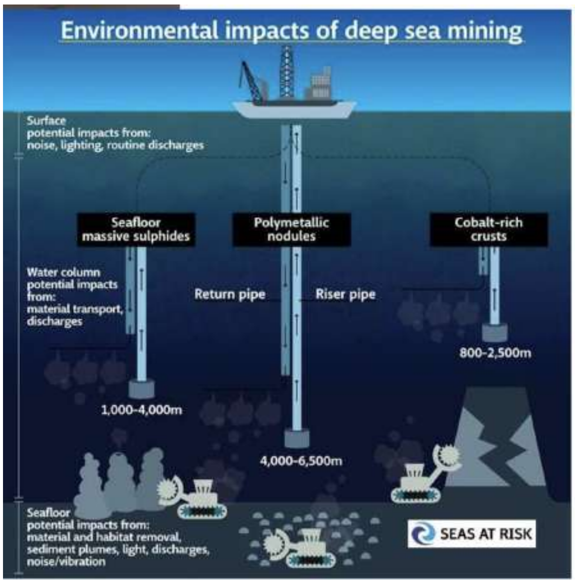


Figure 28: Deep Sea Mining

19. Future Impact
    - Polymetallic Nodule Mining
    - CO_2 Sequestration
    - Climate Change
    - Methane Hydrate Extraction

5 Lecture 19 Hong Kong Marine Biodiversity

1. Hong Kong
   - Most densely populated metropolis in the world
   - Limited efforts on biodiversity management and conservation
   - High environmental variation
   - Low water quality due to human impacts
   - Small marine area : 1651 km^2
   - Long coastline
   - Transition zone between **Palaeartic Japonic** and **Tropical Oriental Regions**
   - Extreme diverse intertidal zone
   - Rocky Shore : Formed by granitic and volcanic promontories with deep wave cut bay
     - Both exposed and sheltered
     - Rocky oysters dominate Mid-shore in sheltered areas
     - Barnacles dominate in Mid-shore in exposed areas
     - Motile Animal in Mid-shore : Chitons, Limpets, Crabs, Snails
       $\Rightarrow$ Mostly grazers and few carnivores
       $\Rightarrow$ E.g. Whelks
     - High Intertidal : Dominate by littorinid snails
     - The lower part of the shore
       (a) Patches of coralline crusts
       (b) Rock Oysters
       (c) Brown Macroalgae
   - Variety of Muddy and Sandy Shore in Hong Kong bays
     $\Rightarrow$ Continental Shelf of Hong Kong : **Muddy Sands**
   - Soft Shore : Sandy Shore $\rightarrow$ Mudflats
     - Muddy flats mostly in Deep Bay and Inner Tolo Harbour

  - Diversity of burrowers
  - Faunal Composition depend on level of wave exposure
  - Nursery grounds for ecologically and commercially important species
- Seagrass : Appears at lower shore to shallow subtidal
  - Sediment stabilizers
  - Support grazers and are feeding grounds for fishers crabs and gastropods
- Corals : In sub-littoral areas
  - Distribution related to dissolved inorganic nitrogen & Dissolved Inorganic Phosphates & Particulate Suspended Matter
  - 84 species Reef Building corals in Hong Kong
  - Some loss of habitat in south Hong Kong due to **eutrophication**
    $\Rightarrow$ Shift to more stress tolerant species

2. Seasonality
   - Influenced by northeast and southwest monsoons (Wind)
   - Temperate : Cold + Dry Winters
   - Tropical : Hot + Wet Summers
3. Coastal Current
   - Upwards current during summer
   - Downwards current during winter
4. Kuroshio Current
   - Influential during winter
   - Bring warm waters to the coast of China

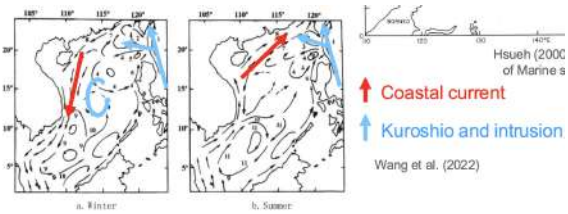


Figure 29: Current Movements

5. Current Effects on the Fauna
    - Temperate Algae is abundant (Only in colder months)
    - Corals persist in Hong Kong due to the warm waters by Kuroshio Current
6. Tides
    - Mixed Tides
    - Diurnal and Semi-diurnal
7. Hong Kong Mangrove
    - 8 mangrove species in Hong Kong
    - Larger Mangroves : West
    - East Mangrove : In Embayments and sandy and stony areas
      ⇒ With smaller trees
      ⇒ E.g. Dwarf Mangroves
8. Hong Kong Marine Biodiversity
    - Close to the Indo-Australian Archipelago Hotspots (Coral Triangle)
    - Coral Triangle : Extremely high marine biodiversity
    - Rich marine biota
      ⇒ 60% Polychaetes found in south china sea in Hong Kong
      ⇒ < 2.7% Sponges
      ⇒ 29% Hermit Crabs
9. Land Reclamation
    - Habitat Destruction

(a) Coastal Habitat
(b) Benthos

- Changes in water quality

(a) Resuspend particulate solid and pollutants
(b) Change hydrodynamics